

Harvesting Healthcare 4.0 : Using Virtual Reality for Enhanced Rehabilitation and AI for Pandemic Prediction



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Enhanced Bi-Manual Rehabilitation for Multiple-Sclerosis using Virtual Reality

People living with Multiple-Sclerosis (MS) may experience functional limitations from disease progression. Rehabilitation in MS involves strategies to improve or maintain function, prevent complications and enhance overall quality of life. In recent years, Virtual Reality(VR) has emerged in the area of physical therapy; several approaches has been taken, from entertaining gaming platforms to larger scale immersive simulators. In this research we are to develop our own Immersive VR rehabilitation system with a reality base derived from hand gestures, sensors and haptics to enhance MS patients' recovery and diagnostics.

Background & Literature Review

Unlike other industries, where the implementation of AR/VR seems to be more of the icing on a cake (most notably in Entertainment), in Healthcare, it is the opposite. VR/AR solutions can make the existing health care system more useful than it ever was . In physical therapy, VR gives the opportunity to provide training and interventions that may not be possible or safe in the real world; it's a means for better understanding of neurological diseases in an entertaining and non-stressful fashion (**Figure 1**). As their illness progresses, patients with MS may suffer from motor impairments, that is why feedback and enhancing recovery for bi-manual gestures are essential for daily activities and that is exactly what we are going to address in our work.



Figure 1- an Overview of VR Rehabilitation [1].

Objectives

Our main goal will be to develop an efficient/flexible system to determine the short- and long-term impact of virtual reality rehab on Bi-manual functions, activity limitation, and MS disease remission rate. We will make sure to assess whether our technology will provide any additional benefits to that of well trained and experienced therapists.

Methodology

Our work will be focused on Bi-manual tasks, we will be using a basic VR headset coupled with some hardware to generate the experiment environment and save some diagnostic useful data (Bi-manual gestures EMG signals). **Figures 2.a and 2.b** previews the type of technologies we will be include in our proof-of-concept setup.

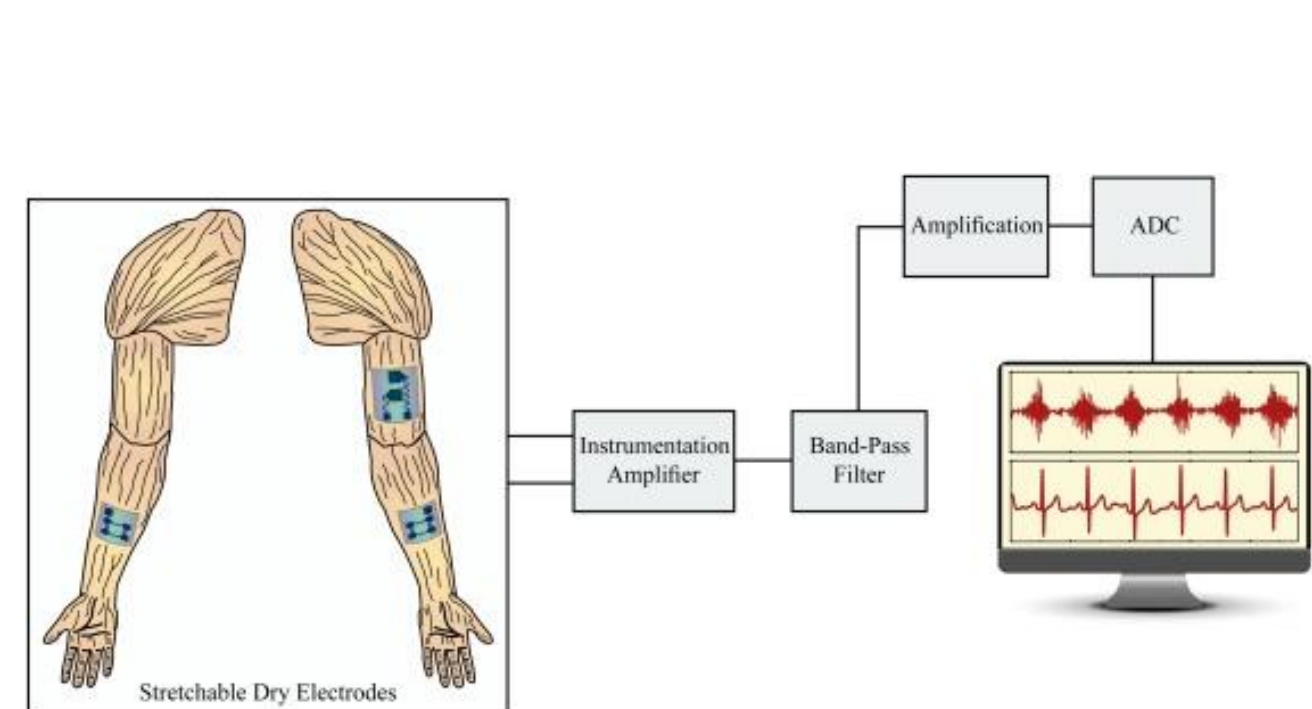


Figure 2- a EMG Data Acquisition.



Figure 2-b Leapmotion Hand Gesture Tracking[2].

We will create a 1st environment as a tutorial, a user-friendly interface for assistance and feeling/reflex assessment with sensors and haptics (**Figure 3.a**), The second target environment will emulate a real-life scenario on a grocery shop simulation (**Figure 3.b**).

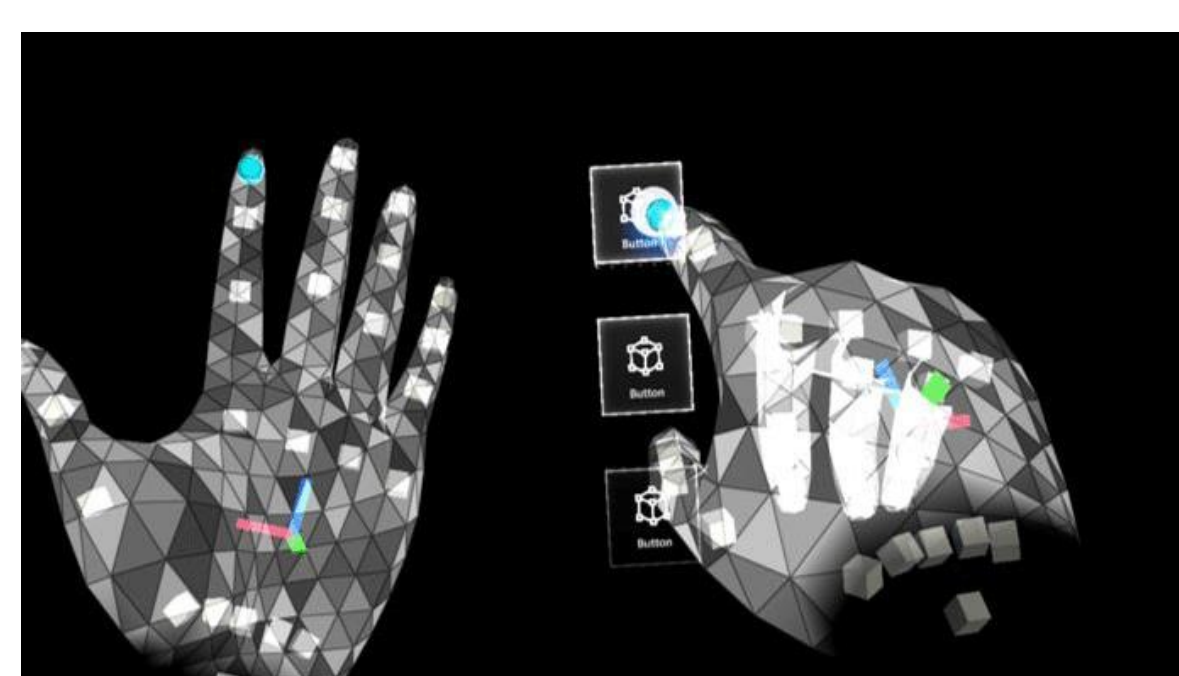


Figure 3-a Microsoft Mixed Reality Home [3]

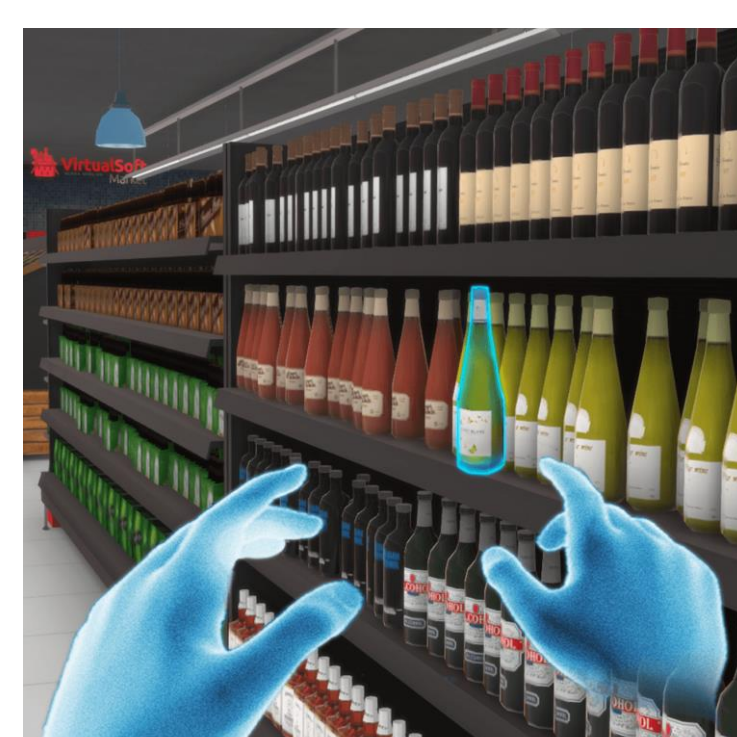


Figure 3-b Steam VR Supermarket [4]

Expected Results & Significance

At the end of the first year, we hope to achieve a deep understanding of the area of the research and of all the techniques that we will need to use to develop a proof of concept that will enhance MS Rehabilitation and will open the door for further investigation on this type of therapy compared the traditional approaches.

Using Wearable Devices Physiological parameters for Covid-19 early Prediction

According to clinical studies [5], the most common symptoms at the onset of COVID-19 are fever, cough, and fatigue, which are closely related to the physiological signs measured by wearable devices. Therefore, it may be a good idea to predict the epidemic trend of COVID-19 using a prediction model based on the wearable device data.

Background & Literature Review

On February 08, 2021, a group of researchers at Mount Sinai found that subtle changes in a participant's heart rate variability (HRV) measured by an Apple Watch were able to signal the onset of COVID-19 up to seven days before the individual was diagnosed with the infection via nasal swab, and to identify those with symptoms [6].

A study performed at the Medical School of Stanford University derived the same conclusion and used retrospectively collected data to develop an online method for potential real-time, early detection of illness onset, the physiological parameters they based their research on are shown in **Figure 4**.

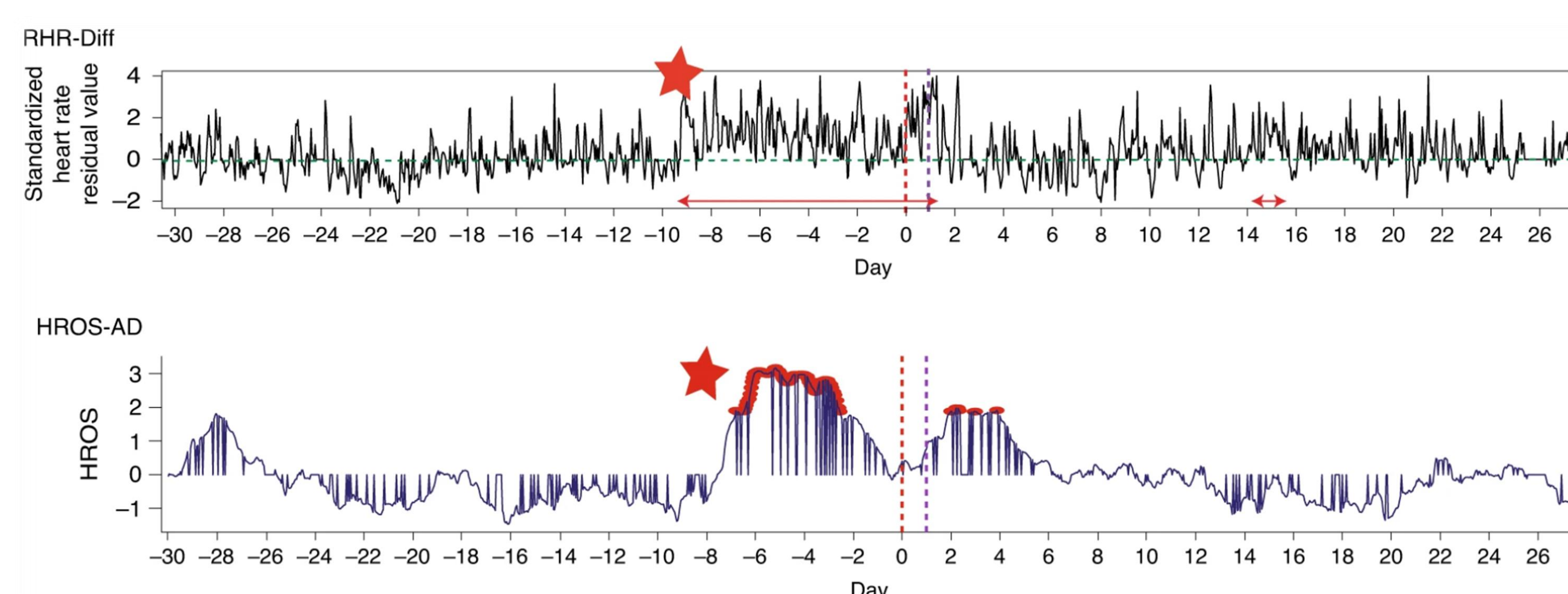


Figure 4- Overview of the derived heart rate metrics used during the study [7]

The data types collected included heart rate, steps and sleep over a period of several months. Two infection detection algorithms were developed (RHR-Diff and HROS-AD), centered around the onset of symptoms (day 0). The earliest detected abnormal heart rate elevations are marked by red stars. The anomaly periods detected by RHR-Diff are spanned by red arrows. The anomaly time points detected by HROS-AD are marked by red dots. The symptom onset day and diagnosis day are indicated by vertical dashed red and purple lines, respectively [7]. The research results showed that better performances when resting heart rate, sleep and activity metrics were combined into a single metric, and that when sensor metrics were combined with self-reported symptoms. This why we intend combine further parameters such as body temperature/humidity or blood saturation related metrics to improve the detection performance even more.

Objectives

Our main goal will be to develop a fast and early prediction model for COVID-19 using non-invasive physiological parameters that are available through wearable devices. The 1st goal will be to select the most appropriate parameters to add to previous studies to get more precise results. We will then conduct different trials on the available datasets to select the detection algorithm we will be running in our study.

References

1. *Virtual Reality for Rehabilitation Overview*, ScienceSoft, McKinney, Texas, "scnsoft.com", accessed 15 Sep 2021.
2. *Leap Motion Hand Tracking Device*, UltraLeap, San Francisco, California "ultraLeap.com", accessed 16 Sep 2021.
3. *Microsoft Mixed Reality*, Microsoft, Redmond, Washington, "docs.microsoft.com", accessed 16 Sep 2021.
4. *Steam VR Supermarket*, Steam, Bellevue, WA, "store.steampowered.com", accessed 15 Sep 2021.
5. Guokang Zhu, "Learning from Large-Scale Wearable Device Data for Predicting the Epidemic Trend of COVID-19", *Discrete Dynamics in Nature and Society*, vol. 2020, Article ID 6152041, 8 pages, 2020.
6. Hirten, Robert P., et al. "Use of physiological data from a wearable device to identify SARS-CoV-2 infection and symptoms and predict COVID-19 diagnosis: observational study." *Journal of medical Internet research* 23.2 (2021): e26107.
7. Mishra, T., Pre-symptomatic detection of COVID-19 from smartwatch data. *Nat Biomed Eng* 4, 1208–1220 (2020).